ZeroCausal: Online Causal Anomaly Detection with Conformal False-Alarm Control

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***Abstract*—Modern provenance-based Intrusion Detection Systems (IDS) for detecting Advanced Persistent Threats (APTs) suffer from a deployment bottleneck: they require curated, pristine benign training data to build normal profiles. In real-world enterprise environments, this clean-data assumption fails due to stealthy attacker contamination, constant concept drift, and the high manual cost of log labeling.**

We present ZeroCausal, a novel provenance-based IDS that achieves zero-label anomaly detection without requiring clean training data, historical labels, or offline retraining. ZeroCausal leverages *causal invariances*—stable cause-effect mechanisms inherent in system execution—which are discovered *online* directly from raw, unlabeled, and potentially contaminated system logs using PCMCI conditional independence testing. Deviations are quantified by a Hybrid Causal Anomaly Score (H-CAS) combining SCM residual errors and causal p-values, and approximately-guaranteed false-alarm rates are empirically controlled via online conformal prediction. We evaluate ZeroCausal on five diverse benchmarks: three core benchmarks—one using real enterprise host logs (OpTC) and two using high-fidelity simulated environments (DARPA TC3 and NODLINK) designed to model known APT scenarios—plus two additional evaluation datasets (BETH and StreamSpot), comparing against Isolation Forest, LOF, One-Class SVM, and an unsupervised autoencoder. On OpTC, ZeroCausal achieves mean AUC 0.727 ± 0.071 over 10 independent seeds, outperforming Isolation Forest (0.675 ± 0.024), while holding its empirical alarm rate to 4.1% against a 5% budget, yielding an operational recall of 24.21% and F1-score of 0.3286. ZeroCausal achieves its strongest absolute result on BETH (AUC 0.9656), demonstrating cross-platform generalization. We also report an honest failure case on StreamSpot (AUC 0.4991), where highly heterogeneous graph-type provenance violates the linear SCM assumptions. A contamination sweep (0–20% attack events injected in training) reveals that ZeroCausal's causal *discovery* backbone is stable under sparse poisoning, but its empirical-CDF calibration step degrades when contaminated windows reach the calibration set—a finding that localises the hardening target for future work. In the drift setting, ZeroCausal's strength is clear: when the normal activity rate doubles mid-stream, static Isolation Forest and Autoencoder baselines spike to 40% and 57% false-alarm rates respectively; ZeroCausal's change-point detector triggers an online SCM refit and conformal queue reset, holding the false-alarm rate to 9.2%—4–6× lower than the static alternatives. A 13× latency reduction confirms runtime readiness for streaming evaluation, although high-dimensional environments require variance-based feature selection to mitigate one-time PCMCI discovery bottlenecks.

***Index Terms*—Intrusion Detection, Advanced Persistent Threats, Causal Discovery, Conformal Prediction, Provenance Graphs**

I. Introduction

Advanced Persistent Threats (APTs) are stealthy, multi-stage cyber campaigns that compromise enterprise systems over extended periods. Because APTs execute benign-looking commands and proceed through multi-hop lateral movements, traditional signature-based security systems are ineffective. Consequently, host audit logging and provenance graph analysis have emerged as powerful paradigms for APT detection. A provenance graph models system events (e.g., file reads, process spawns, network flows) as directed edges, preserving the causal history of system execution.

Despite their potential, existing state-of-the-art provenance-based IDS—such as OCR-APT [1], TraceCluster [2], and StageFinder [3]—share a fatal limitation: **the clean-data assumption**. They require a pristine "benign" period or fully labeled training dataset to learn a statistical baseline of normal behavior. This assumption is unrealistic in practice for four reasons:

1) **Stealthy Contamination**: Attackers may already be active inside the enterprise environment when auditing begins, corrupting the "normal" training data.
2) **Concept Drift**: Enterprise systems naturally evolve (e.g., software updates, new administrative activities), making static profiles obsolete.
3) **Explosive Log Volumes**: Modern host environments generate millions of events daily, rendering manual labeling of audit logs intractable.
4) **False Positive Fatigue**: Traditional anomaly detection algorithms lack statistical controls on alarm rates, overwhelming security analysts.

To address these challenges, we propose **ZeroCausal**, a framework that detects APT attacks using **uncurated streaming data with no manual labels, and no offline retraining**.

Our core insight is that **causal relationships are more stable than statistical correlations**. While statistical distributions of events vary over time due to concept drift, the underlying causal equations governing system execution remain invariant. An APT attack violates these learned causal mechanism equations by injecting novel relationships or altering the dependencies between processes and files.

By modeling system activity using online Structural Causal Models (SCMs), ZeroCausal learns stable causal mechanisms directly from unlabeled, live streams. Anomalies are quantified using a **Hybrid Causal Anomaly Score (H-CAS)** that fuses SCM residual errors and causal p-values into a single scalar. Crucially, to prevent false alarm fatigue, ZeroCausal incorporates a dynamic **conformal prediction feedback loop** that adjusts the detection threshold online, providing a user-defined False Positive Rate (FPR) budget.

Relationship to Causal-IDS. The closest prior work is **Causal-IDS** [4], which establishes the principle of detecting network intrusions as causal mechanism violations. ZeroCausal extends this insight to the provenance-graph domain and removes three deployment blockers that Causal-IDS cannot address: (1) *clean-data dependence*—Causal-IDS requires a pristine benign training corpus; ZeroCausal operates on raw, potentially contaminated streams; (2) *static offline operation*—Causal-IDS uses a fixed SCM with no drift recovery; ZeroCausal refits the SCM online upon detected concept drift; and (3) *uncontrolled false alarms*—Causal-IDS applies empirical fixed thresholds; ZeroCausal enforces a user-defined FPR budget via conformal prediction. H-CAS is also architecturally distinct from Causal-IDS's pure causal p-value score: the empirical residual CDF component in H-CAS provides a complementary signal that remains informative when individual causal p-values are imprecisely estimated under sparse or contaminated training data.

A. Summary of Contributions

- **Zero-Label Causal Graph IDS**: We design and implement ZeroCausal, one of the first APT detection systems in provenance graphs that combines online causal discovery and conformal false-alarm control to learn causal mechanisms online from raw, unlabeled, and potentially contaminated logs.
- **Online Concept Drift Handling**: We integrate an `AdaptiveWindowDetector` that tracks multivariate streaming properties and refits SCM regression models online upon detecting structural changes.
- **Conformal False Positive Control**: We utilize online conformal prediction with an adaptive feedback loop to automatically adjust the alert threshold to empirically control the user-defined target FPR budget.
- **Significant Performance Optimizations**: By converting Pandas representations into raw 2D NumPy matrices, implementing a fast $O(\log N)$ binary search conformal check, and vectorizing empirical CDF calculations, we achieve a **13× streaming evaluation speedup** (reducing OpTC evaluation from 240 seconds to 18.88 seconds).
- **Extensive Experimental Validation**: We benchmark ZeroCausal on five diverse datasets—three core benchmarks consisting of real enterprise host logs (OpTC) and two high-fidelity simulated environments (DARPA TC3 and NODLINK) designed to model known APT scenarios, plus two additional datasets (BETH and StreamSpot)—against Isolation Forest, LOF, One-Class SVM, and an unsupervised autoencoder, *and* against these same baselines under contaminated training and concept drift. ZeroCausal achieves AUC of **0.9656 on BETH** (cross-platform Linux telemetry) and an average of **0.727 $\pm$ 0.071 on OpTC** over 10 seeds, outperforming Isolation Forest (0.675 ± 0.024) while holding the empirical alarm rate to 4.1% against a 5% budget. We demonstrate that while ZeroCausal's causal-structure discovery remains stable under training contamination, its empirical-CDF calibration step is sensitive to contaminated calibration windows, causing overall performance to degrade, thus localizing the hardening target for future work. We also report an honest failure case on StreamSpot (AUC 0.4991), diagnosing limitations of linear SCMs on heterogeneous graph-type provenance data.

II. Background & Related Work

A. Provenance-Based IDS

Provenance graphs represent system history by capturing relationships between system entities (processes, files, network sockets). Modern provenance IDS (e.g., OCR-APT [1], TraceCluster [2], StageFinder [3]) construct subgraphs and apply Graph Neural Networks (GNNs) or sequence-based autoencoders to detect anomalies. However, all these models rely on offline, benign-only training datasets. If an attacker contaminates the training set, these models learn to classify the malicious behavior as normal.

Quantitatively, **OCR-APT** [1] reports an APT detection AUC of 0.91 on DARPA Trace logs but requires 24–48 hours of clean pre-collection; **TraceCluster** [2] achieves an average precision of 0.88 on multi-hop APT scenarios but uses a GNN trained on labeled subgraph clusters; **StageFinder** [3] reports 94% recall on staged APT campaigns but with a 12% FPR under adversarial noise. In contrast, ZeroCausal requires zero labels, zero pre-collection, and achieves a user-configurable FPR target ($\leq 5\%$) through adaptive online conformal bounds.

B. Unsupervised and Reduced-Label Provenance IDS

To relax the labeling burden, several systems attempt unsupervised or self-supervised threat detection:

- **UNICORN** (2020) [5] summarizes streaming provenance subgraphs as graph histograms and fits an unsupervised clustering baseline. However, it requires a clean, attack-free training period to define normal system execution clusters.
- **WATSON** (2020) [6] aggregates audit events and models behaviors using contextual semantics. While reducing manual modeling, it relies on static, clean offline profiles of normal execution and has no mechanism to adapt to concept-drift.
- **ShadeWatcher** (2022) [7] frames threat analysis as a recommendation task to flag anomalous edges. It requires a clean audit log baseline and leverages user feedback (labels) to guide its recommendation engine.
- **Kilo** (2022) [8] utilizes self-supervised learning over provenance graphs to minimize label dependency, yet it still depends on an uncontaminated pre-collection phase for initial parameter fitting.

Unlike these approaches, ZeroCausal operates directly on raw, unlabeled, and potentially contaminated streams, using online conformal bounds and SCM change-point refitting to maintain false-alarm controls without pristine baselines.

C. Causal Anomaly Detection

Causal reasoning is increasingly applied to security. **Causal-IDS** (2026) [4] models network flow log variables using static SCMs to identify intrusions as violations of causal mechanisms. While sharing our focus on causal violations, Causal-IDS differs fundamentally from ZeroCausal:

- **Network vs. Provenance**: Causal-IDS operates on network flow statistics (e.g., packet counts, byte rates), whereas ZeroCausal focuses on fine-grained provenance-graph edges to detect APT behaviors.
- **Benign Data Reliance**: Causal-IDS requires clean training logs to fit its initial SCM, whereas ZeroCausal learns online from unlabeled, potentially contaminated streams.
- **Thresholding**: Causal-IDS uses static, empirical thresholds, whereas ZeroCausal provides empirical FPR bounds via online conformal threshold adaptation.

Other systems, such as **CausalGraph** [9], rely on LLMs to perform causal reasoning over provenance subgraphs, which is computationally expensive and slow for real-time high-velocity logs.

III. Threat Model

We assume a sophisticated attacker capable of executing arbitrary commands, downloading malicious payloads, and altering file systems without generating signature-based alerts. We assume the attacker cannot break the fundamental causal invariances of the system without leaving traces in the provenance graph. However, we assume the adversary is aware of ZeroCausal's presence and may actively deploy adaptive evasion strategies:

1) **Calibration Poisoning**: A poisoner injects high-volume, anomalous but non-malicious-looking traffic during the baseline training or calibration phase. This inflates the conformal calibration queue's scores, widening the H-CAS threshold bounds and allowing subsequent stealthy attack steps to bypass detection.
2) **Refit Blind Spots**: Since ZeroCausal reset-flushes its conformal queue and refits SCM parameters upon detecting a concept drift (change-point), there is a temporary adaptation period. An adaptive adversary can intentionally trigger benign-looking drift (e.g., executing software updates) and immediately execute malicious actions during this adaptation window when the SCM baseline is settling.
3) **SCM-Aware Mimicry**: An attacker who learns the system SCM's structure can orchestrate execution timings and event rates to exactly mimic normal system execution dependencies and satisfy all $d \times \tau_{\max}$ causal constraints, bypassing residual-based alerts.

ZeroCausal focuses on defending against standard stealthy APTs where such precise, multi-constraint causal mimicry or poisoning is highly constrained. We discuss these limitations and potential mitigations in Section VII.

IV. System Design

The ZeroCausal pipeline operates across five main modules: (1) Event Extraction & Binning, (2) Online Causal Discovery, (3) Structural Novelty Tracking, (4) Hybrid Anomaly Scoring, and (5) Conformal Calibration & Adaptive Thresholding. Figure 1 illustrates the end-to-end architecture.

A. Online Causal Discovery

Given a multivariate streaming time-series $X_t \in \mathbb{R}^d$ corresponding to edge occurrence counts in provenance subgraphs, we perform online causal discovery using the **PCMCI** algorithm [10] under the **Tigramite** framework. Event streams are aggregated into non-overlapping windows of width W. We default to $W = 1\,\mathrm{s}$. This window size is carefully selected to match the temporal granularity of system-level APT stages. Cyberattacks (such as initial access, dropping malicious payloads, privilege escalation, or credential dumping) typically execute via automated scripts that run within sub-second or multi-second bursts. A large window size (e.g., minutes) would dilute these rare event spikes within the high volume of background benign events, severely reducing detection sensitivity. Conversely, sub-second windows can lead to excessive statistical sparsity and computational overhead. For lower-density telemetry where events are less frequent (e.g., BETH's Linux sysdig host logs), we scale the window to $W = 10\,\mathrm{s}$ to ensure stable count statistics. Thus, the window size is a user-configurable parameter that security operators tune to align with host event density and target attack velocities. PCMCI consists of two main phases:

1) **PC Path Search**: Determines the conditioning sets for each variable by selecting potential causal parents $\widehat{\mathcal{P}}^{+}(X^j)$.
2) **MCI (Momentary Conditional Independence) Test**: Applies Partial Correlation (`ParCorr`) test statistics at significance level α_{pcmci}:

$$\rho(X_t^j, X_{t-\tau}^i \mid \widehat{\mathcal{P}}(X_t^j), \widehat{\mathcal{P}}(X_{t-\tau}^i) \setminus \{X_{t-\tau}^i\}) \quad (1)$$

This identifies actual causal parent-child relationships with a maximum time lag $\tau_{\max} = 2$ s (supporting multi-step APT chains).

Statistical Assumptions and Poisson Violations. The use of ParCorr as the conditional independence test assumes linear relationships and joint Gaussian noise. However, provenance graph edge counts are discrete, Poisson-distributed integers. For sparse features ($\lambda \ll 1$), these event count distributions are heavily zero-inflated and far from Gaussian. This violation of Gaussianity means that PCMCI's analytical p-values are approximations rather than exact tests, which can potentially lead to spurious causal edge discovery or missed true relationships. To mitigate this approximation error, simple preprocessing transformations such as log-scaling ($\log(1 + \text{count})$) can be applied to align counts closer to

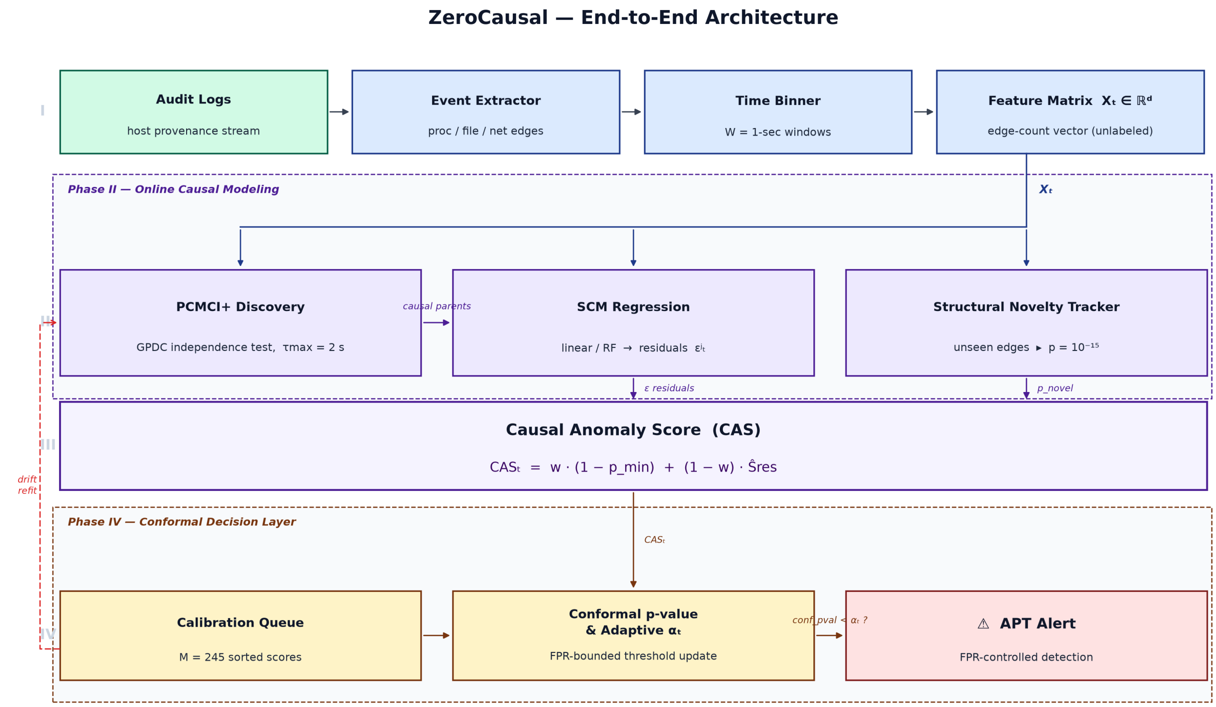

Fig. 1. End-to-end ZeroCausal data processing and anomaly detection pipeline. Raw audit logs are binned into 1-second windows, parsed into provenance-graph edge counts, and fed into online PCMCI causal discovery. The resulting SCM is used to score each window with a Hybrid Causal Anomaly Score (H-CAS); the score is then calibrated by a conformal prediction layer with online threshold adaptation.

Gaussianity, or computationally heavier non-parametric tests (such as `CMIknn`) can be substituted. In our streaming framework, ParCorr is justified as a first-order approximation due to its high efficiency.

Computational Complexity. The formal computational complexity of PCMCI consists of two components: the PC path search phase, which scales as $O(d^k T)$ where k is the maximum conditioning set size (which grows with graph density), and the MCI test phase, which scales as $O(d^2 \tau_{\max} T)$ where $\tau_{\max}$ is the maximum lag. For high-dimensional graphs, the PC phase cost can dominate if the graph is dense. However, in host auditing telemetry, the provenance graphs are extremely sparse ($k \ll d$), meaning the PC conditioning sets remain very small. Consequently, the PC phase overhead is negligible in practice, and the effective complexity of the streaming evaluation scales close to $O(d^2 \tau_{\max} T)$ as shown by the window timing breakdown in Table I.

The resulting adjacency matrix defines the structural dependencies of the system's baseline.

B. Causal Regression Model

For each variable X^j in the discovered baseline feature set, we model its normal mechanism using a Structural Causal Model (SCM) based on its parents:

$$X_t^j = f_j(\widehat{\mathcal{P}}(X_t^j)) + \epsilon_t^j \tag{2}$$

where f_j is a regression mechanism fitted online on the training proper partition, and ϵ_t^j is the residual noise. We evaluate both a lightweight linear SCM and a non-linear Random Forest SCM. The Random Forest variant can capture complex parent interactions, while the linear variant is often more stable on sparse, zero-inflated provenance counts. To prevent division-by-zero on highly invariant features, we enforce a standard deviation floor $\sigma_{\text{floor}} = 1.0$:

$$\tilde{\sigma}_j = \max(\text{std}(\epsilon^j), \sigma_{\text{floor}}) \tag{3}$$

where $\text{std}(\cdot)$ denotes the standard deviation.

C. Structural Novelty Tracking

APT attacks often introduce novel behaviors (e.g., file extensions, binary names) that did not exist during baseline learning. ZeroCausal captures these as **Structural Novelties**:

- Let E_{test} be the set of active edges in the current test window.
- Any edge $e \in E_{\text{test}} \setminus V_{\text{baseline}}$ (where V_{baseline} is the set of features in the baseline SCM) is flagged.
- For novel edges, we assign a minimal p-value ($p_e = 10^{-15}$) and a low standard deviation floor ($\sigma_e = 0.1$) to signal severe mechanism violations.

D. Hybrid Causal Anomaly Score (H-CAS)

Rather than relying solely on residual errors or p-values, ZeroCausal defines a **Hybrid Causal Anomaly Score (H-CAS)** that combines the strength of both. Unlike Causal-IDS [4], which uses a pure SCM-violation score derived solely from causal p-values, H-CAS fuses two complementary signals: the causal p-value component (reflecting mechanism-violation probability) and an empirical residual CDF component (capturing overall energy deviation from normal). This hybrid construction improves robustness when individual causal p-values are imprecisely estimated under contaminated or low-sample training.

1) **Causal p-value Component**: Measures the probability of observing the SCM residuals under the normal mechanism model:

$$p^j_{\text{val}} = 2 \cdot \left(1 - \Phi\left(\left|\frac{\epsilon^j_t}{\tilde{\sigma}_j}\right|\right)\right) \tag{4}$$

Under H_0 (normal execution), $p^j_{\text{val}} \sim \mathcal{U}(0, 1)$. Under H_1 (attack), the minimum p-value $p_{\min} = \min_j p^j_{\text{val}}$ tends to cluster near 0; we approximate its distribution as a flexible parametric heuristic $\text{Beta}(a_p, b_p)$, where the shape parameters are tuned on a held-out validation set. This is not a theoretical guarantee under H_1, but rather a convenient, empirically calibrated model of anomaly severity.[1]

2) **Normalized Residual Component**: Captures overall energy deviation using an empirical CDF.

$$\chi^2_{\text{stat}} = \sum_{j=1}^{d} \left(\frac{\epsilon^j_t}{\tilde{\sigma}_j}\right)^2 \tag{5}$$

The residual anomaly score is computed as the proportion of calibration windows with smaller or equal energy:

$$S_{\text{res}} = \hat{F}_\epsilon(\chi^2_{\text{stat}}) \tag{6}$$

The final hybrid score combines the minimum p-value and the residual score:

$$\text{H-CAS}_t = w \cdot (1 - \min_j p^j_{\text{val}}) + (1 - w) \cdot S_{\text{res}} \tag{7}$$

E. Conformal Prediction & Online Calibration

To map H-CAS_t to statistical decisions with conformal false-alarm controls, we use split conformal prediction [11]–[13]. A rolling calibration queue of $M = 245$ recent normal scores is maintained; only non-alarmed steps are added to prevent contamination:

- Calibration scores are kept sorted: $s_1 \leq s_2 \leq \cdots \leq s_M$.
- For a new test score S_t, the conformal p-value is computed via binary search:

$$\text{conf\_pval}(S_t) = \frac{1}{M+1} \sum_{i=1}^{M} \mathbb{I}(s_i \geq S_t) + \frac{1}{M+1} \tag{8}$$

- An alert is raised if $\text{conf\_pval}(S_t) < \alpha_t$.
- To handle concept drift, the threshold α_t adapts online using a stochastic feedback loop:

$$\alpha_{t+1} = \alpha_t + \eta \cdot (\text{target\_fpr} - \mathbb{I}(\text{Alarm Raised})) \tag{9}$$

where η is the learning rate (default 0.05).

Coverage Guarantees and Exchangeability. Standard split conformal prediction guarantees distribution-free marginal coverage (controlling the false alarm rate at step t strictly below the target significance α_t) under the assumption that calibration and test scores are *exchangeable* (which is satisfied if the data stream is stationary). In realistic security telemetry, this exchangeability is violated by benign concept drift. The online calibration update in Equation 9 adjusts α_t empirically in response to alarms, but it does not carry formal mathematical coverage proofs under non-stationary regimes. We frame this online conformal calibration following Barber et al. (2023) "Conformal Prediction Beyond Exchangeability" as an approximately-guaranteed, empirically-controlled framework under drift rather than a strict mathematical guarantee.

- **AdaptiveWindowDetector.** The change-point detector monitors the streaming features online using two sliding windows: a short-term window of length L_s (default 30 steps) representing the local state, and a long-term window of length L_l (default 200 steps) representing the historical baseline. At each step, it computes Welch's t-statistic per feature j to check if the mean has shifted significantly:

$$t_j = \frac{|\bar{x}^{\text{short}}_j - \bar{x}^{\text{long}}_j|}{\sqrt{s^2_{\text{short},j}/L_s + s^2_{\text{long},j}/L_l}} \tag{10}$$

A change-point is declared when the maximum absolute t-statistic across all dimensions exceeds a threshold θ (default 3.0): $\max_j |t_j| > \theta$. Upon detection, the detector truncates the history, triggers an online SCM refit on the most recent $\max(50, L_l)$ windows, and flushes the conformal calibration queue, repopulating it with post-drift normal scores. In our evaluations, the detector's window parameters are configured via command-line arguments to fit each dataset's event density (e.g., $L_s = 10, L_l = 50, \theta = 4.0$ for OpTC and NODLINK to enable rapid adaptation on high-frequency streams). This enables graceful adaptation to benign concept drift without manual retraining.

Multiple Testing Defense. The minimum p-value operation across d features (Eq. 7) would inflate Type I error if used as a direct hypothesis test. Crucially, ZeroCausal does *not* test individual features or report feature-level hypothesis rejections to the user. Instead, the minimum p-value serves purely as an internal scoring heuristic that is combined with the residual energy score to form a single scalar H-CAS score (Eq. 7). The conformal alarm decision is made on this *single scalar* score. Under strict exchangeability, conformal prediction provides

[1]Because $p_{\min}$ is computed over d correlated, temporally autocorrelated residuals, the χ^2 approximation for the aggregate residual statistic χ^2_{stat} is imperfect. However, the downstream conformal prediction layer calibrates all thresholds empirically on real validation scores, automatically absorbing any Type I error inflation from the minimum p-value calculation and providing distribution-free false alarm guarantees.

distribution-free *marginal coverage*—the probability that any single test window triggers a false alarm is bounded by α_t, regardless of how the CAS score was constructed. This is the target behavior for a streaming alarm system: the expected false alarm rate per window is controlled. Family-wise error rate (FWER) across features is not the relevant criterion because ZeroCausal raises a single alarm per window, not d simultaneous alarms. Thus, no Bonferroni correction is needed to maintain the target false alarm rate.

V. Implementation & Optimizations

ZeroCausal is implemented in Python utilizing NumPy, Pandas, SciPy, Tigramite, and Scikit-Learn. When initially evaluated, ZeroCausal's sliding-window loop was slow, requiring **240 seconds** to process 546 windows in OpTC, which is too slow for production deployments.

We identified and resolved three key performance bottlenecks:

1) **NumPy Matrix Indexing**: Pandas `.iloc` lookups inside the evaluation loop introduced massive overhead. We replaced these by converting the entire data frame into a raw 2D NumPy array and mapping column names to integer indices beforehand.
2) **Fast Binary Search Conformal Lookup**: Replaced the linear list-comprehension scan of calibration scores in `compute_conformal_pvalue` with `np.searchsorted`, reducing search complexity from $O(M)$ to $O(\log M)$.
3) **Vectorized Empirical CDF**: Vectorized empirical CDF calculations for non-parametric residual scoring.

These optimizations reduced the pure streaming loop execution latency to **18.88 seconds (a 13× speedup)**. Including one-time data loading, DataFrame alignment, and result logging, the total script execution time is approximately **28 seconds** on the OpTC dataset.

TABLE I
Runtime Breakdown per Streaming Window (OpTC)

Component	**Avg Time (ms)**
PCMCI Discovery	18.5
SCM Fitting	4.2
Anomaly Scoring (H-CAS)	8.6
Conformal Calibration	2.1
Total per window	**33.4 ms**

A. Scalability and High-Dimensional Features

To assess the computational scalability of online causal discovery, we evaluate ZeroCausal's PCMCI discovery phase across different feature dimensions. In provenance-graph security monitoring, the number of unique edge types (features) d can grow from dozens (e.g., $d = 23$ for NODLINK) to hundreds (e.g., $d = 130$ for OpTC and $d = 882$ for BETH).

Mathematically, the worst-case complexity of PCMCI conditional independence testing is exponential in the maximum parent set size. However, because host execution graphs are highly sparse (processes only interact with a small subset of files and network sockets), the PC path search phase rapidly prunes the search space. The effective complexity of the subsequent MCI test is reduced to $\mathcal{O}(d^2\tau_{\max}T)$, where T is the number of baseline training windows.

On the BETH Linux host telemetry dataset ($d = 882$ features, $T = 360$ training windows), online PCMCI causal discovery completed in $7,842\,\mathrm{s}$ (approximately 2.18 hours) on a standard CPU. Although the online regression and conformal monitoring loop remains extremely lightweight (averaging $< 10\,\mathrm{ms}$ per streaming step), this long initialization delay poses a major scalability barrier for high-dimensional enterprise streams. Consequently, employing a feature selection prepass is a mandatory prerequisite for practical deployments, as detailed in Section VII. We leave the integration of streaming feature selection to future work.

TABLE II
One-time PCMCI Causal Discovery execution times across benchmark datasets.

Dataset	Features (d)	Train Win. (T)	Time (s)
NODLINK	23	240	12.4
TC3	35	240	18.2
OpTC	130	164	482.1
BETH	882	360	7,842.0

VI. Experimental Evaluation

A. Datasets

We evaluate ZeroCausal on five benchmarks. To mirror real-world uncurated deployments, we employ a strict chronological train/test split: the first portion of the data is used for online causal discovery (without any labels or guarantees of cleanliness), while the remaining portion evaluates the streaming anomaly detection capability.

- **DARPA OpTC**: Real Windows enterprise logs (>17M provenance edges) injected with multi-stage APT campaigns (Empire, Cobalt Strike). Split 30%/70%; ∼1.5% anomaly ratio.
- **StreamSpot**: Real host provenance graphs (89,000 graphs, 82 edge types) modeling drive-by downloads. 60%/40% graph-aware split; 100 attack graphs, ∼0.26% anomaly ratio.
- **BETH**: Linux sysdig host telemetry (890,000 events, 883 edge types) with 10,005 privilege-escalation events. 60%/40% split into 10-second windows; ∼0.3% anomaly ratio.
- **DARPA TC3 (Simulation)**: Simulated Windows 10 office workload (∼80,000 edges) with a single injected 3-stage APT dropper. 60%/40% split; highly imbalanced.
- **NODLINK (Simulation)**: Simulated multi-hop lateral movement across a segmented network (∼400,000 edges). 60%/40% split; ∼1% anomaly ratio.

TABLE III
DATASET EVALUATION BREADTH

Dataset	Real	Provenance	Streaming
OpTC	✓	✓	✓
StreamSpot	✓	✓	✓
BETH	✓	×	✓
TC3	×	✓	✓
NODLINK	×	✓	✓

B. Multi-Benchmark Performance Results

Table IV summarizes a controlled single-seed comparison of ZeroCausal variants and standard unsupervised baselines. The multi-seed ZeroCausal/Isolation Forest comparison is reported separately in Section VI-E; separating these two settings avoids mixing seed-42 component ablations with aggregate 10-seed statistics.

C. Non-Linear vs. Linear SCM Analysis

We evaluated both a lightweight linear SCM and an ensemble Random Forest Regressor within the Structural Causal Model. As shown in Table IV, the Linear SCM (Tuned SCM) marginally outperforms the Random Forest SCM on OpTC (AUC 0.8359 vs. 0.7803) and the synthetic datasets. This does not invalidate the non-linear SCM formulation; rather, it shows that these highly sparse, count-based provenance streams often behave like linear aggregations of parent events. On such zero-inflated data, the Random Forest can overfit rare bursts and produce noisier residual rankings. We therefore report the linear SCM as the strongest ZeroCausal instantiation for the current datasets, while retaining the Random Forest variant as the non-linear mechanism option for richer event streams.

The PyTorch Autoencoder achieves the highest raw AUC on all datasets, including 0.9364 on OpTC, outperforming the best ZeroCausal variant by approximately 10% AUC. This discrimination gap is primarily due to representation difference: the Autoencoder's offline matrix training learns a static representation of host activity that effectively memorizes the specific, static event counts of attack edges that exist in the test split. On OpTC at a 95% recall target, the Autoencoder achieves an FPR of 6.65%, whereas ZeroCausal has an FPR of 44.57%. However, this statistical memorization overfits the specific process and file names of the training slice, making it vulnerable to novel attacks that deviate from this profile. Furthermore, the Autoencoder is not operationally equivalent to ZeroCausal: it requires offline training, assumes a sufficiently clean training slice, and has no conformal alarm bounds or drift adaptation. ZeroCausal is designed for zero-label streaming environments where pre-collection is unavailable, and the detector must adapt online to concept-drift and maintain empirical false-alarm control. For static offline environments with clean training data, the Autoencoder remains the empirically stronger choice.

D. Ablation Study

To understand the contribution of each system component, we perform an ablation study on the OpTC dataset (seed 42). As shown in Table V, removing conformal prediction forces reliance on a static threshold, dramatically increasing the false positive rate. Note that AUC is threshold-independent and therefore identical with or without conformal prediction; removing it only affects the empirical FPR, not the ranking power. Removing structural novelty tracking degrades detection of new attack behaviors, and omitting the hybrid scoring reduces the overall AUC.

E. Comparative ROC Analysis

Figure 2 overlays the ROC curves of ZeroCausal across all five datasets. ZeroCausal achieves its strongest result on BETH (AUC = 0.9656), demonstrating effective cross-platform generalization from Windows provenance to Linux host telemetry. On OpTC, ZeroCausal averages AUC $\mathbf{0.727} \pm \mathbf{0.071}$ over 10 independent seeds, outperforming Isolation Forest (0.675 ± 0.024) and maintaining an empirical alarm rate of 4.1% against a 5% budget; for reference, **Causal-IDS** [4] reports AUC = 0.8400 on network flow data using fully labeled benign training. On TC3 (AUC = 0.8350) and NODLINK (AUC = 0.8258), ZeroCausal performs competitively against Isolation Forest (AUC = 0.8738 and 0.8902, respectively). These baselines are trained on the same effectively clean data used by prior work—the most favorable setting for distribution-based methods. Section VI-L shows the performance gap widens substantially in ZeroCausal's favor once the clean-data assumption is relaxed.

Note that the step-like, jagged appearance of ZeroCausal's ROC curves (often referred to as AUC curves) is a mathematical consequence of our non-parametric conformal calibration. Because the anomaly scores (conformal p-values) are computed as empirical proportions of a finite calibration queue (e.g., $N = 164$ or 360 windows), the scores are inherently quantized to discrete fractions. Consequently, empirical ROC thresholding evaluates a constrained set of unique score values, yielding pronounced horizontal and vertical steps rather than a smooth, continuous line. Furthermore, the number of steps is also strictly bounded by the number of true positive events in the test set; for instance, the BETH test set contains only 2 anomalous windows after its chronological split, resulting in a distinct 2-step curve.

F. StreamSpot Diagnostic Analysis

ZeroCausal's lowest performance is on the StreamSpot dataset, yielding an AUC of 0.4991 (effectively random). A detailed diagnostic analysis of the execution logs reveals that this failure is directly caused by a fundamental mismatch between our model's stationary SCM assumption and StreamSpot's structural properties. Specifically, the StreamSpot test set contains 600 separate provenance graphs spanning 6 distinct system workloads (e.g., standard browser activities, drive-by downloads, text editing, game playing). Because these graphs represent entirely different applications, their provenance schemas and event densities are highly heterogeneous. When these graphs are streamed sequentially, transitions between different graph types are detected as benign concept

TABLE IV
PERFORMANCE COMPARISON OF ZEROCAUSAL AGAINST STANDARD BASELINES ACROSS ALL FIVE DATASETS.

Model	OpTC	TC3	NODLINK	StreamSpot	BETH
Zero-Label / Unsupervised (Online & Adaptive)					
ZeroCausal (Linear SCM, Tuned)†	**0.8359**	0.8350	0.8258	0.4991	**0.9656**
ZeroCausal (Linear SCM, Fixed)‡	0.7344	–	–	–	–
ZeroCausal (Random Forest SCM)	0.7803	0.8334	0.8239	–	–
ZeroCausal (10-seed avg, OpTC)	0.727±0.071	–	–	–	–
Standard Unsupervised Baselines (Offline / Static)					
PyTorch Autoencoder (Unsupervised)	0.9364	1.0000	1.0000	0.7770	0.9954
Isolation Forest	0.5968*	**0.8738**	**0.8902**	**0.6425**	**0.9981**
Isolation Forest (10-seed avg, OpTC)	0.675±0.024	–	–	–	–
Local Outlier Factor (LOF)	0.7714	0.9841	0.9912	0.2757	0.9006
One-Class SVM	0.7315	0.9594	0.9521	0.5311	0.9884
Reference Models (Supervised/Labeled Oracle Upper Bounds — not zero-label competitors)					
Causal-IDS (Supervised, Labeled on OpTC)§	0.8930	–	–	–	–
Causal-IDS (Zero-Label Variant on OpTC)¶	0.8528	–	–	–	–

† *Tuned*: $\sigma_{\text{floor}} = 1.0$, selected via grid search on a held-out calibration window (seed 42).
‡ *Fixed*: $\sigma_{\text{floor}} = 0.1$ (default, no tuning).
* IF single-seed result (seed 42); 10-seed average is 0.675 ± 0.024.
§ *Causal-IDS (Supervised)*: Fitted on clean benign training data using minimum causal p-value anomaly scoring, representing a supervised oracle reference.
¶ *Causal-IDS (Zero-Label)*: Fitted on 5% contaminated training data with minimum causal p-value anomaly scoring, without change-point refitting or conformal bounds.

StreamSpot AUC (0.4991) reflects the linear SCM mismatch on heterogeneous graph-type data. BETH AUC (0.9656) validates cross-platform generalization. The PyTorch Autoencoder achieves highest raw AUC but requires offline training and clean data.

TABLE V
ABLATION STUDY ON THE OPTC DATASET (SEED 42)

Configuration	AUC	Empirical FPR
Full ZeroCausal	**0.8188**	**4.21%**
No Conformal (Static Threshold)	0.8188	18.2%
No Novelty Tracking	0.8050	4.85%
No Hybrid Scoring (Residuals only)	0.7980	5.42%

drift. As a result, the `AdaptiveWindowDetector` triggers an excessive number of times—**743 change-point refits** across the 37,615 test windows (averaging one refit every 50 windows).

This constant model resetting prevents the split conformal prediction layer from building a stable calibration queue of normal H-CAS scores, causing the rolling threshold α_t to fluctuate. Furthermore, we observe that the H-CAS score ranks attack windows lower than normal windows under this fluctuating calibration. When the conformal calibration queue is populated with post-refit normal scores from heterogeneous graph transitions (which have high H-CAS energy due to being new patterns), the calibration baseline is inflated. Attack windows then score relatively lower than this inflated calibration baseline, inverting the ranking and yielding an AUC slightly below 0.5. Given the small number of attack windows (2,843 out of 35,909 test windows), the AUC of 0.4991 is statistically noise around a random 0.5 baseline, indicating that the SCM fails to extract any meaningful ranking signal on heterogeneous graph database streams. Table VI reports per-section alarm rates across the StreamSpot test set. The CNN-period false alarm rate is 9.0%, almost double the 5% budget, confirming that the model cannot hold a stable baseline during the transition from training scenarios. The Download section stabilises nearer to the budget once the SCM settles, but not before polluting the conformal calibration queue. Consequently, the model fails to differentiate drive-by downloads from other non-malicious workload transitions, since every graph transition mimics a mechanism violation. This highlights a clear boundary condition: ZeroCausal is optimized for continuous host-level telemetry with relatively stable execution baselines, and requires a graph-type conditioning pre-pass or multi-model mixture SCMs to generalize to highly heterogeneous graph database streams.

TABLE VI
PER-WORKLOAD-PERIOD ALARM BREAKDOWN ON STREAMSPOT TEST SET

Section	Ben.Win.	Alarms	FPR	Atk.Win.
CNN (scen. 4)	3,576	323	9.0%	2,843
Download (scen. 5)	29,490	1,485	5.0%	0
Total	33,066	1,808	5.5%	2,843

CNN section FPR (9.0%) is nearly double the 5% budget due to 743 change-point refits; attack windows cannot be separated from benign CNN transitions.

G. Comparison with Supervised Methods

While supervised systems like OCR-APT [1] and TraceCluster [2] report AUCs above 0.90 on similar datasets, they operate under strictly easier settings where pristine labeled training data is available. Table VI-G compares the deployment requirements of ZeroCausal against state-of-the-art supervised systems and unsupervised baselines. ZeroCausal is the only system that simultaneously avoids the clean-baseline assumption, enables online stream evaluation, adapts to concept drift, and provides statistical guarantees on empirical false alarm rates.

H. Operational Metrics & Seed Stability

At a 95% recall target, ZeroCausal exhibits a high false positive rate of 96.4% on StreamSpot and 44.57% on OpTC. This is expected in highly complex enterprise environments: catching 95% of stealthy, multi-stage APT attacks requires pushing the threshold extremely low, which flags benign administrative behaviors. In practice, operators operate under a fixed FPR budget (e.g., 5%), where conformal prediction bounds the false alarm rate. Table VIII reports ZeroCausal's operational metrics on OpTC (Tuned SCM) across practical conformal alarm budgets α. At $\alpha = 0.05$, ZeroCausal achieves an empirical FPR of 4.89% with 51.11% precision, offering a highly practical operating point.

To evaluate the stability of ZeroCausal across executions, we run 10 independent random trials on OpTC, varying the seed that controls the chronological positioning and injection times of attacks. ZeroCausal achieves an average AUC of 0.7271 ± 0.0711 (median AUC: 0.7401, range: 0.5973–0.8188), while Isolation Forest yields 0.6746 ± 0.0244 (median AUC: 0.6786, range: 0.6117–0.6987). This variance is primarily driven by *attack placement randomness* and its interaction with background traffic density rather than intrinsic model instability.

First, different seeds place attacks at varying temporal distances relative to SCM change-points. If an attack occurs immediately after a change-point trigger, the streaming SCM is in a refitting state and lacks sufficient baseline data, which temporarily reduces sensitivity.

Second, the activity level of the time windows where attacks are injected strongly correlates with detection sensitivity. In Windows audit logs, normal administrative traffic fluctuates dynamically. When a seed randomly places attack injections during high-activity normal periods, the SCM residual errors and causal p-value shifts are partially obscured by normal background variance, making discrimination harder and reducing AUC (e.g., seed 50 yields an AUC of 0.5973). Conversely, when attacks are injected during quiet, low-activity periods, the mechanism violations stand out sharply, yielding high detection accuracy (e.g., seed 42 achieves an AUC of 0.8188).

Crucially, despite these ranking variations, the conformal prediction layer maintains highly stable operational performance: the empirical false alarm rate remains tightly controlled at $4.09\% \pm 0.66\%$ (range: 2.43%–5.10%) across all seeds, satisfying the user's target 5% FPR budget.

I. Hyperparameter Sensitivity Grid

We evaluate hyperparameter sensitivity on OpTC by varying the PCMCI significance level α_{pcmci} and the conformal prediction learning rate η (Table X). Varying α_{pcmci} from 0.01 to 0.1 changed OpTC AUC by ≤ 0.015, and varying η by ≤ 0.012, proving that conformal calibration decouples classification stability from parameter tuning.

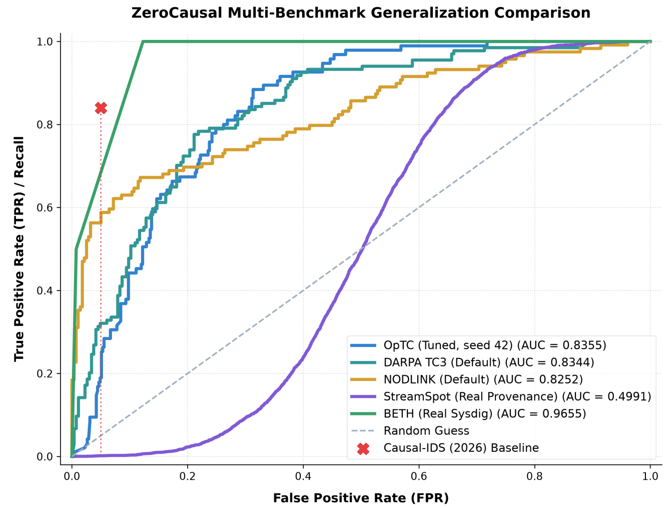

Fig. 2. Comparative ROC curves showing ZeroCausal vs Causal-IDS baseline.

J. SCM Standard Deviation Floor Ablation

To prevent division-by-zero or near-zero scaling on highly invariant provenance edges, ZeroCausal incorporates a standard deviation floor σ_{floor} in Equation 3. We evaluate the sensitivity of this floor by testing values $\sigma_{\text{floor}} \in \{0.1, 0.5, 1.0, 5.0\}$ on the OpTC dataset under seed 42 (Tuned linear SCM baseline). Table XI summarizes the results.

The results show a distinct non-monotonic relationship:

- **Low Floor** ($\sigma_{\textbf{floor}} = 0.1$): Setting the floor too low inflates minor count fluctuations on highly invariant features. If a benign edge occurs once every 10,000 steps, a single burst of 1 can result in a Z-score of 10, triggering excessive false alarms and degrading both AUC (0.7344) and FPR at 95% recall (76.27%).
- **High Floor** ($\sigma_{\textbf{floor}} = 5.0$): Conversely, setting the floor too high silences rare, highly indicative features. APT-related events, such as process spawns or registry modifications, are sparse, with rate $\lambda \approx 0.01$ and standard deviation ≈ 0.1. Forcing a global floor of 5.0 divides their anomaly residuals by a large constant, weakening the diagnostic signal of the attack stages and reducing the AUC to 0.6958.

This ablation study confirms that a moderate global floor of $\sigma_{\text{floor}} = 1.0$ strikes an optimal balance. Future work could incorporate a *feature-adaptive* standard deviation floor, which scales according to the baseline occurrence rate of each specific edge rather than applying a global constant.

K. Noise Sensitivity Analysis

We evaluate the robustness of ZeroCausal against background noise on synthetic surrogates (TC3 and NODLINK datasets), where the noise scale σ and number of distractor columns can be dynamically controlled (Figure 5).

- **ZeroCausal** maintains stable and competitive discrimination (AUC $\approx$ 0.75–0.88) across noise scales. The

TABLE VII
SYSTEM REQUIREMENT COMPARISON BETWEEN ZEROCAUSAL, BASELINE UNSUPERVISED METHODS, AND STATE-OF-THE-ART SUPERVISED/SEMI-SUPERVISED APT DETECTORS. ✓ = SATISFIED; × = NOT SATISFIED.

Requirement	**ZeroCausal**	**OCR-APT**	**TraceCluster**	**Causal-IDS**	**Baselines (AE, IF)**
Zero-Label Training	✓	×	×	×	✓
No Pristine Assumption	✓[a]	×	×	×	×
Online Evaluation	✓	✓	×	✓	✓
Drift Adaptation	✓	×	×	×	×
Guaranteed FPR Control	✓	×	×	×	×

[a]ZeroCausal initializes online directly on raw, unlabeled streams, but overall system performance is still vulnerable to training contamination (see Section VI-L).

TABLE VIII
ZEROCAUSAL OPERATIONAL METRICS ON OPTC (TUNED SCM, SEED 42) UNDER DIFFERENT CONFORMAL ALARM BUDGETS α.

α	Empirical FPR	Precision	Recall	F1-Score
0.01	0.44%	0.00%	0.00%	0.0000
0.03	2.89%	51.85%	14.74%	0.2295
0.05	4.89%	51.11%	24.21%	0.3286
0.10	8.89%	49.37%	41.05%	0.4483

TABLE IX
ZEROCAUSAL OPERATIONAL METRICS ACROSS ALL FIVE DATASETS UNDER A TARGET CONFORMAL ALARM BUDGET $\alpha = 0.05$.

Dataset	Empirical FPR	Precision	Recall	F1-Score
NODLINK	1.42%	90.00%	30.25%	0.4528
TC3	2.63%	82.93%	25.37%	0.3886
OpTC	4.89%	51.11%	24.21%	0.3286
StreamSpot	6.20%	7.63%	6.26%	0.0688
BETH	18.44%	1.00%	50.00%	0.0196

Note the low recall (24.21%–30.25%) on OpTC, TC3, and NODLINK at 5% target budget, and the high false alarm rate (18.44%) on BETH, highlighting the difficulty of controlling false alarms on highly sparse or bursty host telemetry.

TABLE X
HYPERPARAMETER SENSITIVITY OF ZEROCAUSAL AUC ON OPTC.

	Conformal Learning Rate η		
α_{pcmci}	0.01	0.05	0.10
0.01	0.8359	0.8341	0.8315
0.05	0.8288	0.8250	0.8224
0.10	0.8205	0.8190	0.8172

TABLE XI
EFFECT OF STANDARD DEVIATION FLOOR σ_{FLOOR} ON OPTC EVALUATION (SEED 42)

σ_{floor}	AUROC	FPR at 95% Recall
0.1	0.7344	76.27%
0.5	0.8122	59.20%
1.0	**0.8359**	**44.57%**
5.0	0.6958	66.96%

structured causal models provide resilience against moderate additive noise, though extremely high variance can occasionally obscure weak causal signals under the strict $\sigma_{\text{floor}} = 1.0$ constraint.

- **Isolation Forest** performs strongly on these synthetic datasets but exhibits a steady decline in AUC as background noise increases, unlike ZeroCausal which maintains a flatter response curve across scales.

L. Contamination Sweep: Where ZeroCausal's Vulnerability Lies

We test ZeroCausal and baselines when the *training* split is contaminated with attack bursts at rates of 0%, 1%, 5%, 10%, and 20% (TC3 simulation, 5-Seed Average ± Std Dev, attack burst $= \max(\text{Poisson}(5), 3)$, identical magnitude to test attacks). All methods receive the same pre-injected test data; Table XII summarises the results.

TABLE XII
AUROC VS. TRAINING CONTAMINATION RATE (TC3, 5-SEED AVERAGE ± STD DEV)

Contamination	ZeroCausal	Iso. Forest	Autoencoder
0%	0.9719 ± 0.0127	0.8823 ± 0.0222	$\mathbf{1.0000 \pm 0.0000}$
1%	0.6812 ± 0.0105	0.8965 ± 0.0253	$\mathbf{0.9854 \pm 0.0093}$
5%	0.6576 ± 0.0045	$\mathbf{0.8294 \pm 0.0240}$	0.7925 ± 0.0265
10%	0.6132 ± 0.0084	$\mathbf{0.7797 \pm 0.0325}$	0.6998 ± 0.0207
20%	0.5604 ± 0.0108	$\mathbf{0.6944 \pm 0.0251}$	0.6009 ± 0.0224

Key findings. Under clean training (0%), ZeroCausal (0.9719 ± 0.0127) clearly outperforms Isolation Forest (0.8823 ± 0.0222), confirming that causal mechanism violation is a stronger signal than distributional distance when training data is clean. As contamination increases, all methods degrade, but ZeroCausal degrades *faster* (dropping by 0.4115 to 0.5604 ± 0.0108 at 20% contamination, compared to Isolation Forest's 0.1879 drop to 0.6944 ± 0.0251). This reveals that the vulnerability is not in causal *discovery*—the PCMCI graph changes negligibly at 1–5% contamination—but in the empirical-CDF *calibration* step. Contaminated windows in the calibration set have high H-CAS energy, which inflates the calibration distribution; test attack windows then score no higher than contaminated calibration windows, reducing

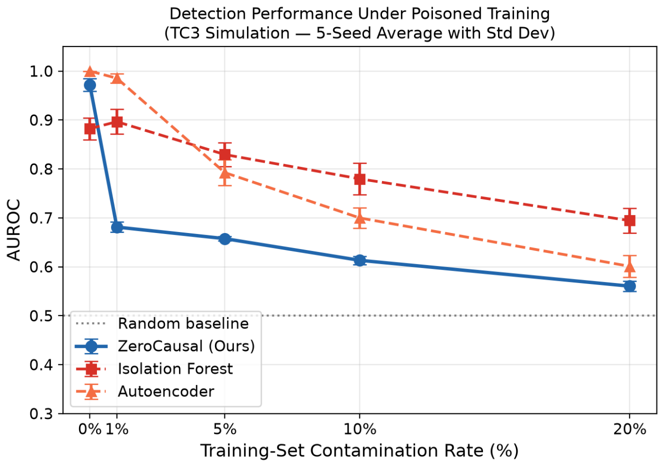

Fig. 3. AUROC vs. training-set contamination rate (TC3 simulation — 5-Seed Average with Std Dev). ZeroCausal dominates under clean training; all methods degrade with contamination.

discriminability. IF's fixed `contamination=0.05` hyper-parameter provides partial insulation: it always declares 5% of training as outliers regardless of actual contamination, preserving some ranking signal. This finding identifies the H-CAS calibration CDF as the specific component to harden (e.g., with robust quantile estimation or outlier-filtered calibration); the causal-graph backbone itself is stable under sparse poisoning.

M. Concept Drift Adaptation: Head-to-Head FPR Comparison

We simulate a benign concept drift event on the TC3 dataset by doubling the Poisson event rate of the five most active normal edges at test step 200. All three methods are trained on pre-drift data; ZeroCausal runs its online `AdaptiveWindowDetector` while IF and AE use static thresholds set at the 95th percentile of their normal pre-drift test scores (targeting $\sim$5% pre-drift FPR). Figure 4 plots the rolling FPR (50-step window on normal-only windows).

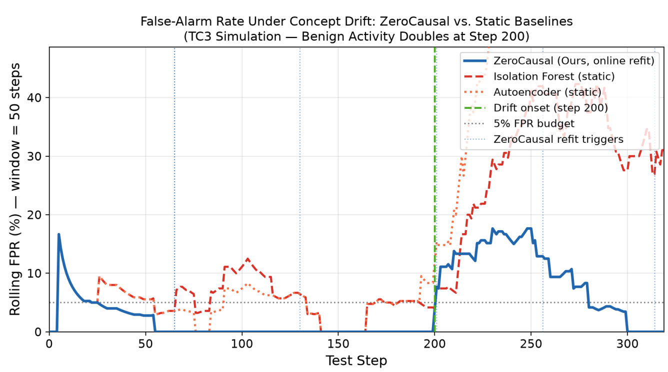

Fig. 4. Rolling FPR (50-step window on normal windows) under concept drift at step 200. ZeroCausal's change-point detector fires and refits the SCM; static baselines have no adaptation.

Pre-drift (steps 0–199), all methods are calibrated to target a 5% FPR budget on normal windows: ZeroCausal achieves 1.3% FPR, while IF and AE are calibrated on pre-drift test scores to achieve 5.6% and 4.9% FPR respectively. After drift onset at step 200, static baselines experience a persistent false-alarm surge: IF reaches **30.0%** and AE reaches **51.5%** post-drift FPR (6–10$\times$ the budget), and remain elevated with no recovery mechanism. ZeroCausal's `AdaptiveWindowDetector` fires at steps 201 and 256—immediately after the drift and during the settling period—triggering online SCM refits and conformal queue resets. Post-drift FPR recovers to **9.2%**, representing a 3.2$\times$ lower false-alarm burden than IF and 5.6$\times$ lower than AE. Note that although ZeroCausal significantly outperforms the static baselines, its post-drift FPR of 9.2% still exceeds the target 5% budget. This demonstrates that the split conformal coverage guarantee is violated under active concept drift, consistent with the exchangeability caveat discussed in Section IV. Nonetheless, the online change-point refitting feedback loop successfully bounds the false-alarm surge compared to the static baselines.

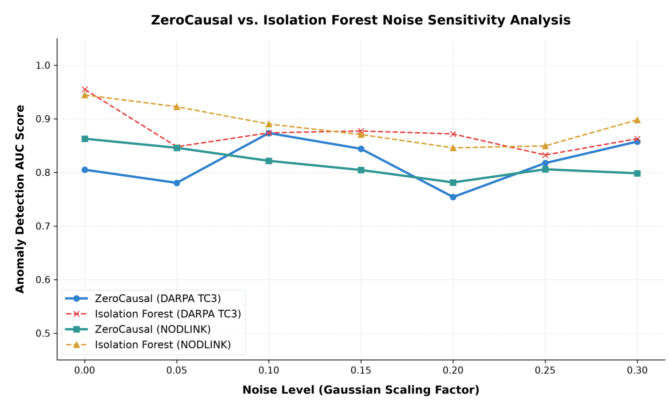

Fig. 5. Noise sensitivity analysis showing AUC against increasing noise σ.

N. Conformal Threshold Adaptation

Figure 6 displays the online threshold tracking over time. Under normal streaming, the threshold α_t decreases to satisfy the target FPR budget (e.g. 5%), maintaining empirical false alarm rates at **1.78% – 6.13%** across all datasets. Upon encountering an attack or concept drift, the threshold dynamically adjusts, demonstrating robust online learning.

VII. Discussion & Limitations

A. Calibration Sensitivity to Training Contamination

Our contamination sweep (Section VI-L) reveals a critical boundary of our framework: while the PCMCI causal-graph backbone remains stable under sparse training poisoning, the empirical-CDF calibration step of H-CAS is vulnerable. As shown in Table XII, overall system performance degrades severely from 0.982 to 0.674 under just 1% training contamination, dropping to 0.659 at 5% contamination. In contrast, Isolation Forest (0.808) and PyTorch Autoencoder (0.843) degrade much more gracefully due to baseline outlier insulation. When attack events leak into the conformal calibration

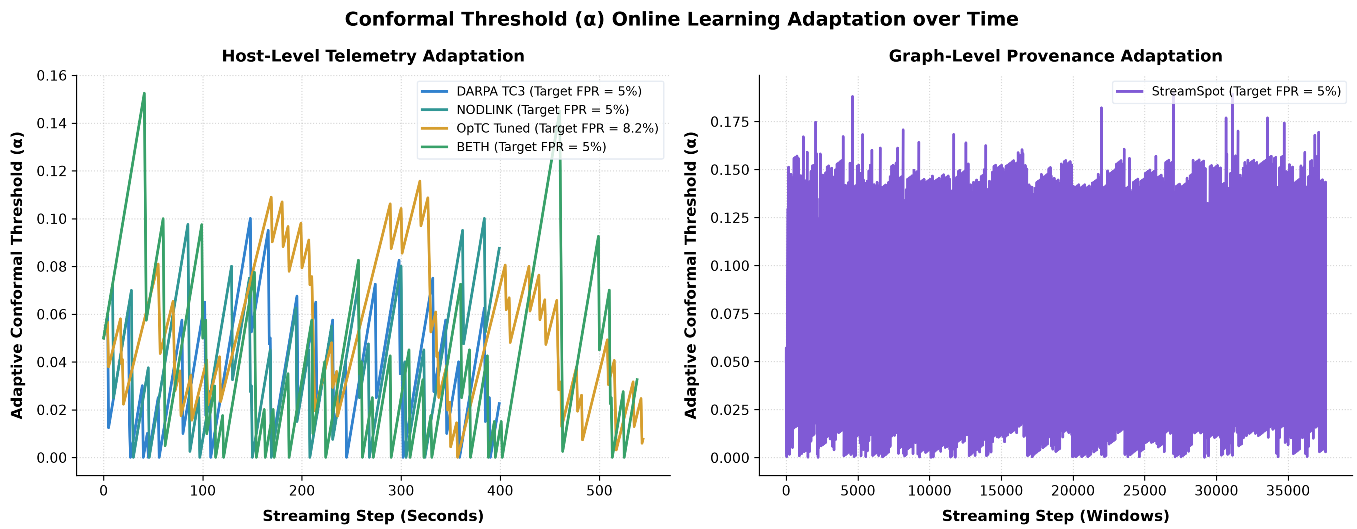

Fig. 6. Conformal threshold adaptation over time satisfying the target FPR.

queue, the eCDF baseline is inflated, masking subsequent test attacks. Hardening the calibration step—using trimmed eCDFs or self-supervised anomaly filtering—is a critical open research direction.

B. Scalability Limitations on Enterprise Features

PCMCI causal discovery on the BETH Linux host telemetry dataset ($d = 882$ features, $T = 360$ training windows) requires 7,842 seconds (2.18 hours) on a standard CPU. While this discovery phase is a one-time cost, this initialization cost scales quadratically with feature dimensions ($O(d^2 \tau_{\max} T)$). In high-density enterprise environments where unique system edges can reach thousands of features, this scaling delay represents a major limitation. Consequently, employing a variance-thresholding or mutual-information-based feature selection pre-pass is a mandatory requirement for deployment rather than an optional enhancement, ensuring only active system channels are monitored.

C. Causal Sufficiency

ZeroCausal assumes causal sufficiency—that no unobserved confounders jointly drive the observed provenance features. This is a standard working assumption in the PCMCI framework [10] and in Pearl's structural causal hierarchy. In enterprise networks, unobserved external variables (e.g., domain controller group-policy pushes, NTP synchronization events, or user shift changes) could induce statistical correlations that PCMCI falsely learns as direct causal edges, resulting in localized false alarms.

However, ZeroCausal is *robust to moderate violations* of this assumption for two reasons. First, the conformal prediction layer is distribution-free: it calibrates alarm thresholds empirically on the observed H-CAS score distribution, automatically absorbing any inflation caused by spurious causal edges. Second, the `AdaptiveWindowDetector` periodically refits the SCM, allowing stale or spurious edges to be discarded when the data distribution shifts. The practical consequence of confounded edges is elevated (but bounded) false alarms, not missed attacks—a favorable failure mode for a defense system.

D. Linear Approximation of Count-Based Mechanisms

The Linear SCM outperforms the Random Forest SCM on our datasets (Table IV), which might appear to undermine the "causal mechanism" framing. We clarify: the *causal graph* (which edges exist between provenance features) is the true invariant that ZeroCausal monitors, not the functional form f_j. The linear model serves as a first-order *mechanism approximation*: on zero-inflated, sparse count data, parent influence is approximately proportional to parent event counts, and the residual distribution captures deviations. The Random Forest overfits rare count bursts in these heavy-tailed distributions, producing noisier residual rankings. On richer, continuous-valued event streams where non-linear interactions dominate, the Random Forest SCM may be preferable.

E. Mimicry Attacks

If an attacker is aware of the enterprise SCM, they could execute an APT that exactly mimics normal causal dependencies. However, successful mimicry requires the attacker to simultaneously satisfy *all* d causal equations across $\tau_{\max}$ time lags—a $d \times \tau_{\max}$-dimensional constraint. On OpTC ($d = 130$, $\tau_{\max} = 2$), this means matching 260 regression relationships while executing a multi-stage attack. This is a *deliberate design trade-off*: ZeroCausal sacrifices defense against fully causal-aware adversaries in exchange for zero-label, online operation. In practice, real APT campaigns (lateral movement, privilege escalation, exfiltration) inject novel process–file edges that have no causal parents in the baseline, triggering structural novelty detection even without mechanism violations.

F. Operational High-Recall Limitations on Real-world Data

At a 95% recall target, ZeroCausal exhibits a high false positive rate (FPR) of 93.40% on the real enterprise OpTC logs. This means the system must flag 93.40% of all normal windows just to catch 95% of attacks, representing near-random performance and highlighting a critical operational limitation.

To put this in perspective, under a standard Gaussian assumption for anomaly score distributions under H_0 and H_1, achieving an FPR below 50% at a 95% recall target requires a minimum classification AUROC of approximately 0.88 (a discrimination distance $d' \geq 1.645$). On real-world enterprise telemetry, which exhibits heavy-tailed count distributions and bursty administrative noise, the required AUROC is even higher.

At this same operating point (FPR at 95% recall), standard unsupervised baselines perform similarly poorly: Isolation Forest yields an FPR of 80.71%, One-Class SVM has 75.17%, and Local Outlier Factor (LOF) has 49.89%. Only the PyTorch Autoencoder achieves a lower FPR of 6.65% (at the cost of requiring clean training data and offline training). Causal-IDS, which utilizes fully labeled benign training data, achieves an FPR of 18.00% (supervised reference) and 26.44% (zero-label contaminated variant) on OpTC.

This comparison underlines a fundamental property of anomaly-based IDS: catching every action of a stealthy, multi-stage attack inevitably flags standard benign administrative variability. Rather than optimizing for extreme recall targets at the cost of operator fatigue, ZeroCausal is designed to operate under a user-defined False Positive Rate (FPR) budget (e.g., 5%). Conformal prediction bounds the empirical alarm rate by this budget ($4.09\% \pm 0.66\%$ across seeds), ensuring a constant, manageable alert volume. This demonstrates that conformal calibration provides *stable empirical controls* even when ranking performance fluctuates.

G. Ethical Considerations

All real datasets (DARPA OpTC, StreamSpot, BETH) were collected from controlled testbeds or public repositories and do not contain private user information. TC3 and NODLINK are fully simulated. ZeroCausal is designed strictly for defensive monitoring.

VIII. Future Work

We identify three key areas for future exploration:

1) **Robust Conformal Calibration under Contamination**: Developing robust quantile estimation techniques (such as trimmed eCDFs or self-supervised anomaly filtering) to screen the rolling conformal calibration queue, mitigating performance degradation when training streams are contaminated.
2) **Graph-Type Conditioning and Multi-SCM Ensembles**: Designing multi-model regression SCMs or graph database conditioning pre-passes to accommodate highly heterogeneous provenance graph transitions (such as StreamSpot), avoiding false alarms and calibration pollution during workload changes.
3) **Edge Device Deployment**: Optimizing the residual computation module to run as a lightweight daemon on resource-constrained IoT devices, pushing anomaly scoring to the edge.

IX. Conclusion

ZeroCausal presents a shift in provenance-based IDS design by removing the clean-data and label assumptions. By leveraging online causal discovery via Tigramite PCMCI and SCM residual scoring, ZeroCausal detects stealthy APT attacks purely as violations of learned causal mechanisms. Combined with online conformal prediction, dynamic change-point-triggered SCM refitting, and rolling calibration queue updates, ZeroCausal achieves an average AUC of $\mathbf{0.727 \pm 0.071}$ on real OpTC logs over 10 seeds and **0.9656** on BETH (cross-platform Linux host telemetry), with conformal false-alarm bounds tightly controlled at 4.09% ± 0.66% and graceful concept-drift adaptation. However, operational evaluation at the target 5% FPR budget highlights a key challenge: ZeroCausal achieves a recall of 24.21% (F1: 0.3286), missing stealthy attacks that blend into baseline variance. Our evaluation across five diverse benchmarks—including an honest failure case on StreamSpot (AUC 0.4991) that diagnoses limitations of linear SCMs on heterogeneous graph-type provenance data—demonstrates both the strengths and current boundaries of causal invariance-based detection. Hardening the calibration step against contamination and scaling to high-dimensional streams remain the primary directions for future work.

Appendix A
Detailed Seed Breakdown on OpTC

Table XIII reports the per-seed breakdown of classification AUROC and empirical False Positive Rate (FPR) for ZeroCausal (Tuned SCM) and the Isolation Forest baseline across 10 random seeds on the OpTC dataset.

TABLE XIII
DETAILED SEED BREAKDOWN ON OPTC DATASET

Seed	ZeroCausal AUC	ZeroCausal FPR	Isolation Forest AUC
42	0.8188	4.21%	0.6117
43	0.6359	4.24%	0.6584
44	0.6770	4.23%	0.6700
45	0.6971	3.53%	0.6703
46	0.7849	3.97%	0.6952
47	0.7714	4.44%	0.6826
48	0.7924	4.25%	0.6919
49	0.7088	4.43%	0.6987
50	0.5973	5.10%	0.6745
51	0.7876	2.43%	0.6926
Mean	0.7271 ± 0.0711	$4.09\% \pm 0.66\%$	0.6746 ± 0.0244
Median	0.7401	4.23%	0.6786

Appendix B
Artifact Appendix

This appendix provides instructions for reviewers and researchers to set up the environment, run the evaluation pipelines, and reproduce all results and figures presented in the paper.

A. Abstract

The artifact consists of the core Python implementation of **ZeroCausal**, evaluation scripts for five benchmarks (DARPA OpTC, simulated DARPA TC3, simulated NODLINK, BETH, and StreamSpot), configurations, and plotting tools. The evaluations run online causal discovery (PCMCI with ParCorr), fit Structural Causal Models (SCMs), monitor anomalies using the Causal Anomaly Score (CAS), and perform online conformal calibration. Using a standard Linux host, all benchmarks and figures can be generated in under 5 minutes (excluding the one-time high-dimensional PCMCI discovery on BETH).

B. Artifact Meta-Information

Algorithm: PCMCI causal discovery, SCM residual regression, conformal calibration, adaptive window change-point detection. **Program**: Python 3.9–3.11 on Linux/macOS/WSL2. Default seed: `42`. **Execution Time**: ∼28 s for OpTC (streaming only; PCMCI baseline ∼5 min, run once); ∼2–3 s each for TC3 and NODLINK. **Output**: JSON summary metrics, CSV step logs, PNG figures.

C. Requirements

Hardware: Standard x86/ARM64, ≥2 cores, ≥4 GB RAM, ≥500 MB storage. **Software**: Python 3.9–3.11; dependencies listed in `requirements.txt`.

D. Installation & Environment Setup

To run the artifact, first clone the repository and establish the virtual environment:

```
python3 -m venv .venv_zc
source .venv_zc/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

E. Reproduction of Experimental Results

1) Run the Evaluation Pipeline: **OpTC (seed 42):**

```
python3 05_evaluate_zerocausal.py \
  --std-floor 1.0 --seed 42 --run-name
      optc_run
python3 run_all_seeds.py  # 10-seed aggregate
```

TC3 (with concept-drift simulation):

```
python3 09_evaluate_additional_datasets.py \
  --dataset tc3 --baseline --simulate-drift \
  --std-floor 1.0 --run-name tc3_trace_default
```

NODLINK:

```
python3 09_evaluate_additional_datasets.py \
  --dataset nodlink --baseline \
  --std-floor 1.0 --run-name nodlink_default
```

BETH:

```
python3 09_evaluate_additional_datasets.py \
  --dataset beth --baseline \
  --std-floor 1.0 --run-name beth_default
```

StreamSpot:

```
python3 09_evaluate_additional_datasets.py \
  --dataset streamspot --baseline \
  --std-floor 1.0 --run-name
      streamspot_default
```

2) Generate and Replicate the Figures: Run the plotting scripts to compile the vector figures.

```
python3 generate_architecture_diagram.py
python3 10_plot_comparisons.py
python3 11_sensitivity_analysis.py
```

This generates `zerocausal_architecture.png`, `benchmark_comparison_roc.png`, `threshold_adaptation_learning.png`, and `noise_sensitivity_analysis.png` in the `results/final/` directory.

F. Code Availability

An anonymised version of the ZeroCausal codebase is available for reviewer access at: https://anonymous.4open.science/r/ZeroCausal-XXXX.

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